

PART 0 — How to Read This Paper

A beginner's map for "Attention Is All You Need" (Vaswani et al., 2017)
Lesson companion · original paper also included as a separate PDF

This is Lesson 0 only. Do not rush into architecture equations yet. First learn what a research paper is, how to navigate it, and what this paper is trying to do.

1. What is a research paper?

A research paper is a written argument by researchers that says: **(1)** here is a problem, **(2)** here is our idea, **(3)** here is evidence it works. It is not a textbook chapter. It assumes some background, compresses years of work into a few pages, and is organized for other researchers—not for first-time learners.

That is why first papers feel hard: the writing is dense, notation is compressed, and many ideas are assumed. Your job is not to understand every sentence on the first pass. Your job is to extract the problem, the idea, the method, and the evidence.

2. How to read a machine-learning paper (practical method)

Use multiple passes. Do not try to absorb everything linearly.

Pass 1 — Map (10–15 min): Title, abstract, introduction start, figures, conclusion. Ask: What problem? What claim? What is Figure 1?

Pass 2 — Architecture: Read the model section while staring at the diagram. Ignore perfect math understanding. Ask: input → what happens → output?

Pass 3 — Equations: For each equation, list symbols, shapes, and what computation it performs. Tiny numerical examples help more than rereading.

Pass 4 — Experiments: What datasets? What metric? Better than what baselines? What ablation proves which piece matters?

Pass 5 — Limitations: What did they not solve? What assumptions remain?

3. What problem are the authors trying to solve?

In 2017, the best sequence models (especially for translation) relied heavily on recurrence (RNNs/LSTMs) and sometimes convolutions. Recurrence processes tokens one after another, which makes training hard to parallelize and makes long-range dependencies difficult.

The authors ask: *Can we build a strong sequence transduction model that relies entirely on attention, with no recurrence and no convolution?* Their answer is the Transformer.

4. What does “contribution” mean?

A contribution is the new thing the paper claims to give the field—not just results, but a reusable idea. For this paper, the core contribution is: a new encoder–decoder architecture based solely on attention (self-attention + multi-head attention), showing strong translation quality with better parallelization.

Important distinction: a contribution is what they **introduced and demonstrated**, not every later idea inspired by Transformers (BERT, GPT, Vision Transformers, etc.). Those came after.

5. How to distinguish paper sections

Section type	What it answers	In this paper
Motivation	Why does this matter? What was broken before?	Intro: recurrence is sequential; long-range deps hard; attention helps
Architecture	What is the model? How are pieces arranged?	§3 Model Architecture + Figure 1
Equations	Exact computations (attention, FFN, PE, etc.)	§3.2–3.5 attention, FFN, embeddings, PE
Experiments	How did they test the claim?	§5 Training details; §6 Results
Results	Did it work? By how much?	WMT BLEU gains; speed/parameter comparisons
Limitations	What remains unsolved or fragile?	Often implied; later work clarifies quadratic cost, etc.

6. Mental map of the entire paper

Keep this map in your head. We will fill each box in later parts—without rushing.

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[PROBLEM]
RNN/LSTM translation models are strong but sequential
→ hard to parallelize, hard long-range deps

[IDEA]
Replace recurrence/convolution with attention only
→ Transformer encoder-decoder

[CORE MECHANISM]
Scaled Dot-Product Attention
→ Multi-Head Attention
→ Self-Attention + Cross-Attention + Masked Attention

[SUPPORTING PIECES]
Embeddings + Positional Encoding
Feed-Forward Networks
Residual connections + LayerNorm ("Add & Norm")
Stacked layers (N=6)

[TRAINING]
Teacher forcing, cross-entropy, Adam + LR schedule
label smoothing, dropout, shared embeddings

[EVIDENCE]
WMT EN-DE / EN-FR translation, BLEU scores
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faster training, ablations of components

[WHAT WE WILL BUILD TOWARD]

Input tokens → embeddings+PE → encoder stack

→ decoder (masked SA + cross-attn + FFN)

→ linear + softmax → next-token probabilities

7. How the paper is organized (page-level map)

§1 Introduction — motivation; why recurrence is limiting; claim of attention-only model.

§2 Background — related work (attention, ConvS2S, ByteNet, self-attention precedents).

§3 Model Architecture — the heart: encoder/decoder, attention, FFN, embeddings, PE.

§4 Why Self-Attention — complexity/path-length/parallelization comparison.

§5 Training — data, batches, optimizer, regularization, label smoothing.

§6 Results — translation quality, speed, ablations, English constituency parsing.

§7 Conclusion — summary and future directions.

8. What to open first in the PDF

Open the original paper PDF now. Look only at: (1) the title, (2) the abstract, (3) Figure 1 (the famous architecture diagram), (4) Table 1 or 2 in results if you skim ahead. Do not worry if Figure 1 looks like a machine schematic. By Part 11 you will be able to rebuild it.

Original paper file: [Attention_Is_All_You_Need_Vaswani_2017.pdf](#) (arXiv:1706.03762)

9. Checkpoint before Part 1

Before we learn vectors/matrices/softmax, lock these answers in your own words:

1. What problem is this paper attacking?
2. What is the claimed contribution in one sentence?
3. Which section will contain the architecture vs the experiments?
4. Why are we not starting with equations yet?

Next lesson (when you say so): [PART 1 — only the minimum prerequisites needed to understand this paper.](#)